DSA 107 – Data Science Lab

Lab 1: Environment and First Notebook

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Objective

The purpose of this lab is to set up our data science environment using Google Colab and also GitHub. Furthermore, it aims to practice basic data analysis techniques. By using Python libraries such as matplotlib, pandas and seaborn, we inspected the Palmer Penguins dataset, handled missing values, filtered specific rows, and created a data visualization to check the relationship between body mass and flipper length across different penguin species.

Data Description

In this experiment, we analyzed the Palmer Penguins dataset, which is collected from the Palmer Archipelago in Antarctica. The dataset consists of 344 rows and 7 columns. The columns include both categorical variables such as species, island, sex, and numerical variables such as bill length, bill depth, flipper length and body mass. An analysis of missing values using .isna().sum() revealed missing entries in the dataset. We found that all four physical measurement columns have 2 missing entries, while the sex column has the most missing values with 11 missing rows. This is likely because determining a penguin's sex in the wild is difficult without medical tests.

Methodology

The workflow began by setting up a computing environment in Google Colab and linking it to GitHub for project tracking. Next, the required Python libraries including pandas, numpy, matplotlib and seaborn were imported by running the initial setup cell. The Palmer Penguins dataset was loaded directly into a DataFrame from a remote CSV URL. Then, pandas methods, for instance, .shape, .info() and .describe() were utilized to inspect the dimensions, data types and summary statistics of the data. Missing entries were detected using .isna().sum(). Furthermore, we practiced column selection and applied conditional filtering to analyze species, such as counting Gentoo penguins and calculating their proportion. Finally, a scatter plot was generated using seaborn to visualize flipper length against body mass, categorized by species.

Results

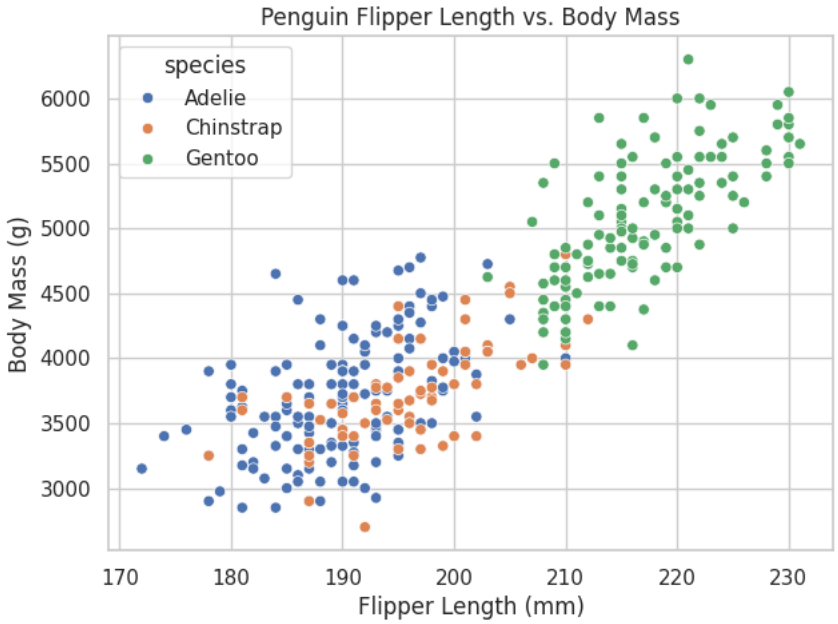


Figure 1: *Penguin flipper length versus body mass categorized by species*

The analysis of the dataset revealed several key findings:

First, conditional filtering showed that there are 124 Gentoo penguins, which make up approximately 36.0% of the total dataset.

Second, the scatter plot in ***Figure 1*** demonstrates a clear positive relationship between flipper length and body mass. As flipper length increases, body mass also increases.

Finally, species similarities are visible in the plot. Adelie and Chinstrap penguins show similar distributions with smaller flipper lengths and body mass, whereas Gentoo penguins are clearly distinguished by longer flippers and higher body mass.

Discussion

The findings show that the positive relationship observed between flipper length and body mass is biologically meaningful. In aquatic birds like penguins, larger body mass requires larger flippers to generate sufficient propulsion and maintain swimming efficiency in cold waters. Furthermore, the clear separation of Gentoo penguins showcases distinct morphological differences across species; Gentoo penguins are naturally larger and adapted to different diving and foraging behaviors compared to Adelie and Chinstrap penguins. However, an important limitation of this simple bivariate analysis is that it does not account for sexual dimorphism. Since male penguins are typically heavier than females and the dataset contains missing sex values, some variance within species clusters remains unexplained.

Reflection

As a student with limited prior experience in Python, especially in the Python libraries, transitioning into this data science environment presented an initial learning curve. Navigating syntax errors, DataFrame indexing, and visualization libraries was challenging at first. To overcome these problems, I utilized Colab's built-in Gemini assistant as a learning aid to understand the logic behind specific functions and debugging steps. While interpreting the generated code and error diagnostics was initially difficult, analyzing each step helped me build an understanding of basic pandas manipulation and Seaborn plotting. As we move forward in the course, I aim to master these foundational programming concepts to conduct analyses with greater independence in upcoming laboratories.